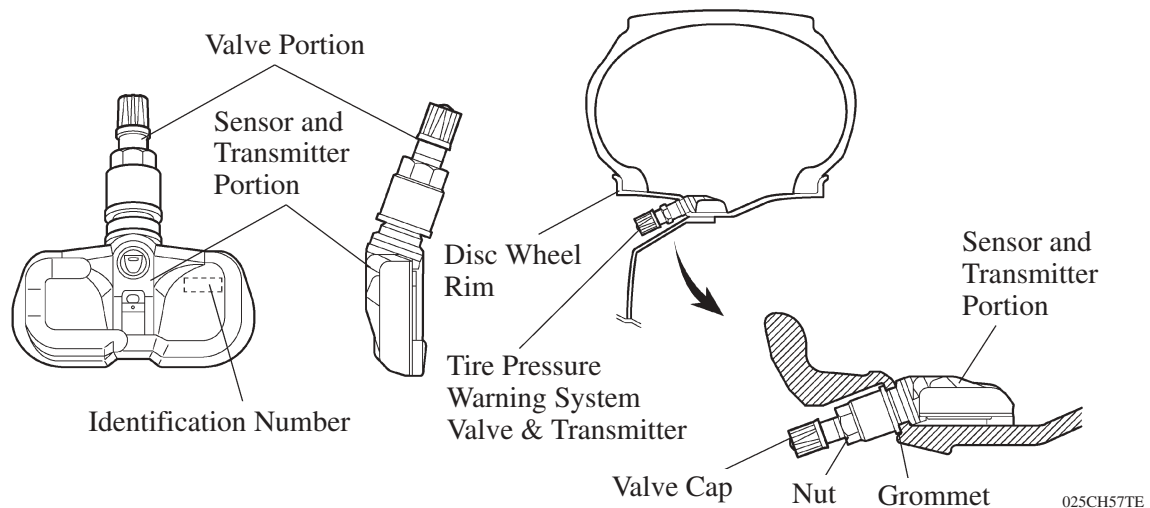


## ■ CONSTRUCTION AND OPERATION

### 1. Tire Pressure Warning System Valve & Transmitter

- The tire pressure warning system valve & transmitter is integrated in the air valve of a disc wheel. It measures the pressure and the temperature of the air in the tire and transmits the measured values and a recognition ID to the tire pressure warning system valve & transmitter.
- If the battery voltage drops, the tire pressure warning system valve & transmitter assembly must be replaced. Furthermore, if the battery voltage drops, the tire pressure warning system valve & transmitter will be unable to transmit signals, which causes a DTC to be output.
- Tire pressure warning system valve & transmitter with 5 different ID code range are used on one vehicle.
- Each tire pressure warning system valve & transmitter has a built-in semi-conductor to directly measure the inflation pressure of the tire.
- Frequency of the tire pressure warning system valve & transmitter is 314.98 MHz.



#### NOTICE

- Ensure the proper direction of the tire pressure warning system valve & transmitter by adhering to the prescribed procedure for installing a tire pressure warning system valve & transmitter on a wheel. Failure to do so could result in an incorrect measurement of the tire air inflation pressure.
- Make sure to replace the tires in accordance with the prescribed procedure. To prevent the tire pressure warning system valve & transmitter from damage, drop the tire pressure warning system valve & transmitter into the wheel before removing the tire. Failure to do so could damage the tire pressure warning system valve & transmitter.

For further details regarding the above, see the 2007 Camry Repair Manual (Pub. No RM0250U).

#### Service Tip

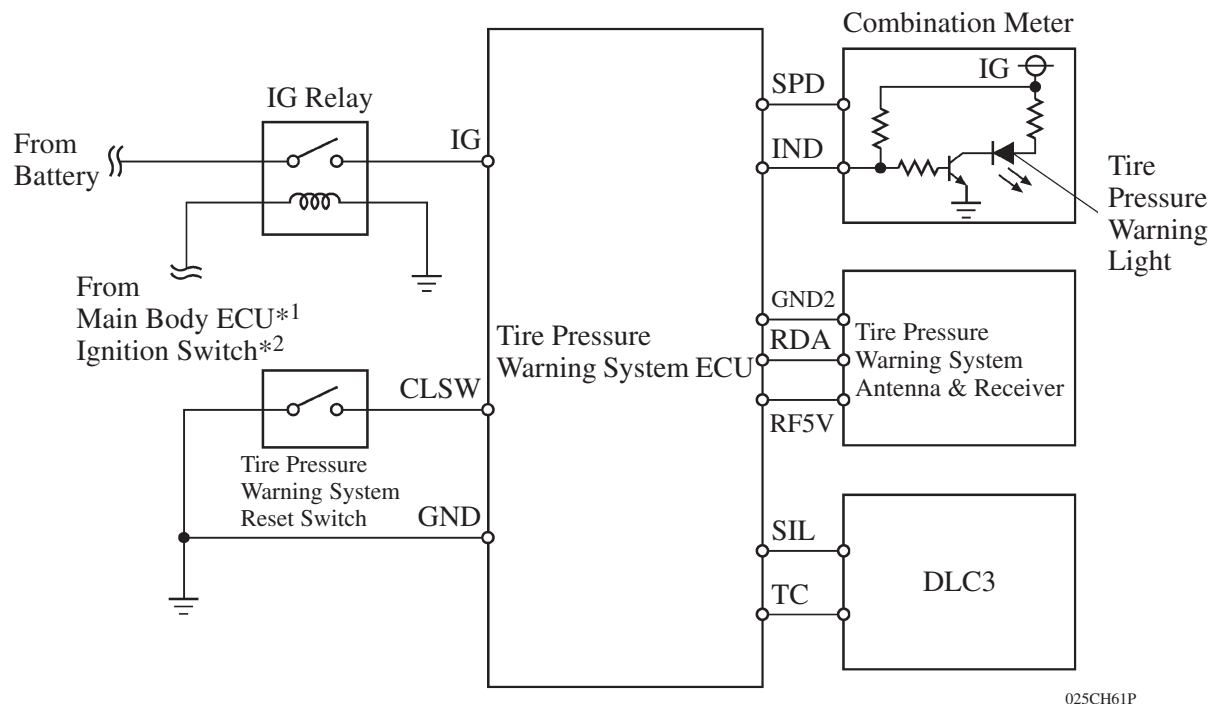
- If the lithium battery is depleted, replace the entire tire pressure warning system valve & transmitter assembly.
- After a tire pressure warning system valve & transmitter has been replaced, the ID of the tire pressure warning system valve & transmitter must be registered in the ECU. To register an ID, use a hand-held tester to enter the ID code that is indicated on the sensor.
- When replacing the tire pressure warning system valve & transmitter, all the separate ID codes of the 5-tire pressure warning system valve & transmitter must be registered. Even if only 1 tire pressure warning system valve & transmitter is replaced, the ID codes of all 5 tire pressure warning system valve & transmitter must be registered again.
- A new tire pressure warning system valve & transmitter that is available as a service part is in the sleep mode in its initial state, to prevent the battery from depleting. During PDS (Pre-Delivery Service), after the tire pressure warning system valve & transmitter and the tire are correctly mounted on the disc wheel, inflating the tire to the specified pressure causes the sleep mode to cancel. Keep the air pressure of the spare tire the same as the set air pressure amount of the front and rear tires.

For further details regarding the above, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

## 2. Tire Pressure Warning System Antenna & Receiver and Tire Pressure Warning System ECU

- The tire pressure warning system antenna & receiver, receives the radio wave signals from the tire pressure warning system valve & transmitter and transmits those signals to the tire pressure warning system ECU.
- The tire pressure warning system antenna & receiver receives the tire inflation pressure data and sensor identification data.
- The tire pressure warning system antenna & receiver can determine from the received data whether the signals came from its own tires.

### ► Wiring Diagram ◀



025CH61P

\*1: Models with smart key system

\*2: Except models with smart key system

### 3. Tire Pressure Warning System Reset Switch

- By operating the tire pressure warning system reset switch, tire pressure warning system ECU can be set to issue a warning at an inflation pressure that corresponds with the type of tires. Therefore, the dealer must set the warning threshold to the proper value in order to comply with the local regulations.
- Operate the tire pressure warning system reset switch only after the inflation pressures of all 5 tires (including the standard type spare tire\*) have been adjusted on the vehicle.
- To initialize the system, press and hold the tire pressure warning system reset switch for 3 seconds or longer with the engine switch turned ON. After the system has been initialized, the tire pressure warning light blinks 3 times at 0.5 Hz.
- During the initialization, the tire pressure warning system valve & transmitter measures the inflation pressure of the tires, and registers the signals that are transmitted into the ECU at a frequency of 1 per minute. The initialization process is completed when the signals from the 5 tires have been received.
- Once the tire pressure warning system reset switch has been pressed, turning OFF the engine switch is not recommended for a few minutes.

\*: Optional Equipment

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#### Service Tip

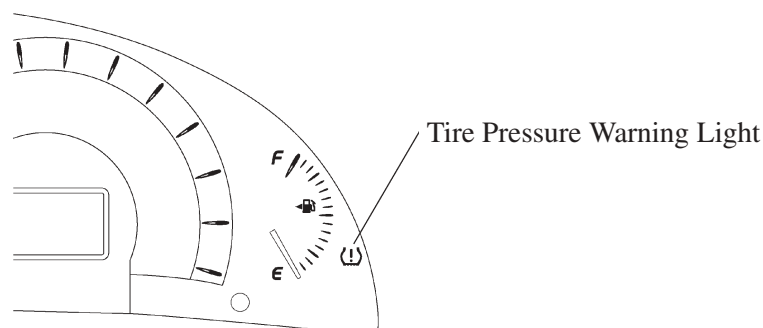
- Since the initialized values are dependent upon the accuracy of the tire pressure gauge to be used, use a tire pressure gauge that has been properly calibrated.
- The system must be initialized in the following conditions:
  - 1) New vehicle delivery
  - 2) Replacement with tires of different size (inflation pressure)
  - 3) Tire pressure warning system ECU replacement

#### CAUTION

If the system is initialized with tires whose inflation pressure deviates from the threshold pressure, the system can be initialized at those values. Therefore, initialize only after the tires have been inflated to the specified air pressure.

### 4. Tire Pressure Warning Light

- The tire pressure warning light is located in the combination meter.
- This warning light illuminates or blinks in accordance with signals from the tire pressure warning system ECU if the vehicles own tires are inflated with low pressure or if malfunction occurs in the system.



## 5. Self-Diagnosis

If malfunctions are detected in the system, the tire pressure warning system ECU warns the driver by illuminating the tire pressure warning light after blinking it at 1 Hz intervals for 1 minute, and stores the DTC (Diagnostic Trouble Code) in the memory.

The DTC that are stored by the tire pressure warning system ECU can be accessed by connecting the SST (09843-18040) to the DLC3 terminals TC and CG, and reading the blinking of the tire pressure warning light. They can also be accessed by connecting a hand-held tester. The table below lists the DTC that pertain to this system. For further details, see the 2007 Camry Repair Manual (Pub No. RM0250U).

### Service Tip

After the DTC have been detected, it is necessary to identify the wheel that contains the faulty tire pressure warning system valve & transmitter. It can be identified by performing the operation in accordance with the prescribed procedure. For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U)

| DTC No.  | Detection Item                                                                            | DTC Deletion Condition                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C2111/11 | Tire pressure warning system valve & transmitter ID1 operation stop (sleep mode)          | When data is received from a transmitter with an ID code that is registered in the ECU, or the ID code of the tire pressure warning system valve & transmitter is newly registered in the ECU. |
| C2112/12 | Tire pressure warning system valve & transmitter ID2 operation stop (sleep mode)          |                                                                                                                                                                                                |
| C2113/13 | Tire pressure warning system valve & transmitter ID3 operation stop (sleep mode)          |                                                                                                                                                                                                |
| C2114/14 | Tire pressure warning system valve & transmitter ID4 operation stop (sleep mode)          |                                                                                                                                                                                                |
| C2115/15 | Tire pressure warning system valve & transmitter ID5 operation stop (sleep mode)          |                                                                                                                                                                                                |
| C2121/21 | Data from tire pressure warning system valve & transmitter registered to ID1 not received | When a DTC deletion operation is implemented or the ID code of the tire pressure warning valve & transmitter is newly registered in the ECU.                                                   |
| C2122/22 | Data from tire pressure warning system valve & transmitter registered to ID2 not received |                                                                                                                                                                                                |
| C2123/23 | Data from tire pressure warning system valve & transmitter registered to ID3 not received |                                                                                                                                                                                                |
| C2124/24 | Data from tire pressure warning system valve & transmitter registered to ID4 not received |                                                                                                                                                                                                |
| C2125/25 | Data from tire pressure warning system valve & transmitter registered to ID5 not received |                                                                                                                                                                                                |
| C2141/41 | A malfunction in the tire pressure warning system valve & transmitter registered to ID1   |                                                                                                                                                                                                |
| C2142/42 | A malfunction in the tire pressure warning system valve & transmitter registered to ID2   |                                                                                                                                                                                                |
| C2143/43 | A malfunction in the tire pressure warning system valve & transmitter registered to ID3   |                                                                                                                                                                                                |
| C2144/44 | A malfunction in the tire pressure warning system valve & transmitter registered to ID4   |                                                                                                                                                                                                |
| C2145/45 | A malfunction in the tire pressure warning system valve & transmitter registered to ID5   |                                                                                                                                                                                                |
| C2165/65 | Abnormal temperature inside ID1 tire                                                      |                                                                                                                                                                                                |
| C2166/66 | Abnormal temperature inside ID2 tire                                                      |                                                                                                                                                                                                |
| C2167/67 | Abnormal temperature inside ID3 tire                                                      |                                                                                                                                                                                                |
| C2168/68 | Abnormal temperature inside ID4 tire                                                      |                                                                                                                                                                                                |

(Continued)

| DTC No.  | Detection Item                                                                                             | DTC Deletion Condition                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| C2169/69 | Abnormal temperature inside ID5 tire                                                                       | When a DTC deletion operation is implemented or the ID code of the tire pressure warning valve & transmitter is newly registered in the ECU. |
| C2171/71 | Tire pressure sensor ID not registered                                                                     |                                                                                                                                              |
| C2176/76 | Tire pressure warning system antenna & receiver is error                                                   |                                                                                                                                              |
| C2177/77 | Initialization incomplete                                                                                  |                                                                                                                                              |
| C2181/81 | Data from tire pressure warning system valve & transmitter registered to ID1 not received (test diagnosis) | When deletion conditions of the ID codes have been established or departing from the test mode.                                              |
| C2182/82 | Data from tire pressure warning system valve & transmitter registered to ID2 not received (test diagnosis) |                                                                                                                                              |
| C2183/83 | Data from tire pressure warning system valve & transmitter registered to ID3 not received (test diagnosis) |                                                                                                                                              |
| C2184/84 | Data from tire pressure warning system valve & transmitter registered to ID4 not received (test diagnosis) |                                                                                                                                              |
| C2185/85 | Data from tire pressure warning system valve & transmitter registered to ID5 not received (test diagnosis) |                                                                                                                                              |
| C2191/91 | Vehicle speed signal error (test diagnosis)                                                                |                                                                                                                                              |